

## **Regfile**

Regfiles Constitute of Address decoder, Basic Register, Multiplexer.

### **Objective:**

The purpose of the regfile is to write and read the data from the defined registers and to generate the other functionality like interrupt.

Based on the input address to the regfile, the address is decoded. If the request is to write the data, the data is written to the register whose address corresponds to that of input address. If the request is to read the data, data is read from the register with the address same as input address.

### **Address Decoder:**

The purpose of the address decoder is to enable the appropriate enable signal based on the address.

Address decoder decodes the address of the registers. Based on the input address, decoder enables the enable signal of the corresponding address register. The address input to the decoder is given by software.

### **Basic register:**

The basic register performs all basic functionality like write or read the data from or to the register.

The enable signal from the decoder, write and read signals and data input from software are the inputs to the basic register.

Based on the enable signal, enabled for particular register, the data is written or read to or from that Register.

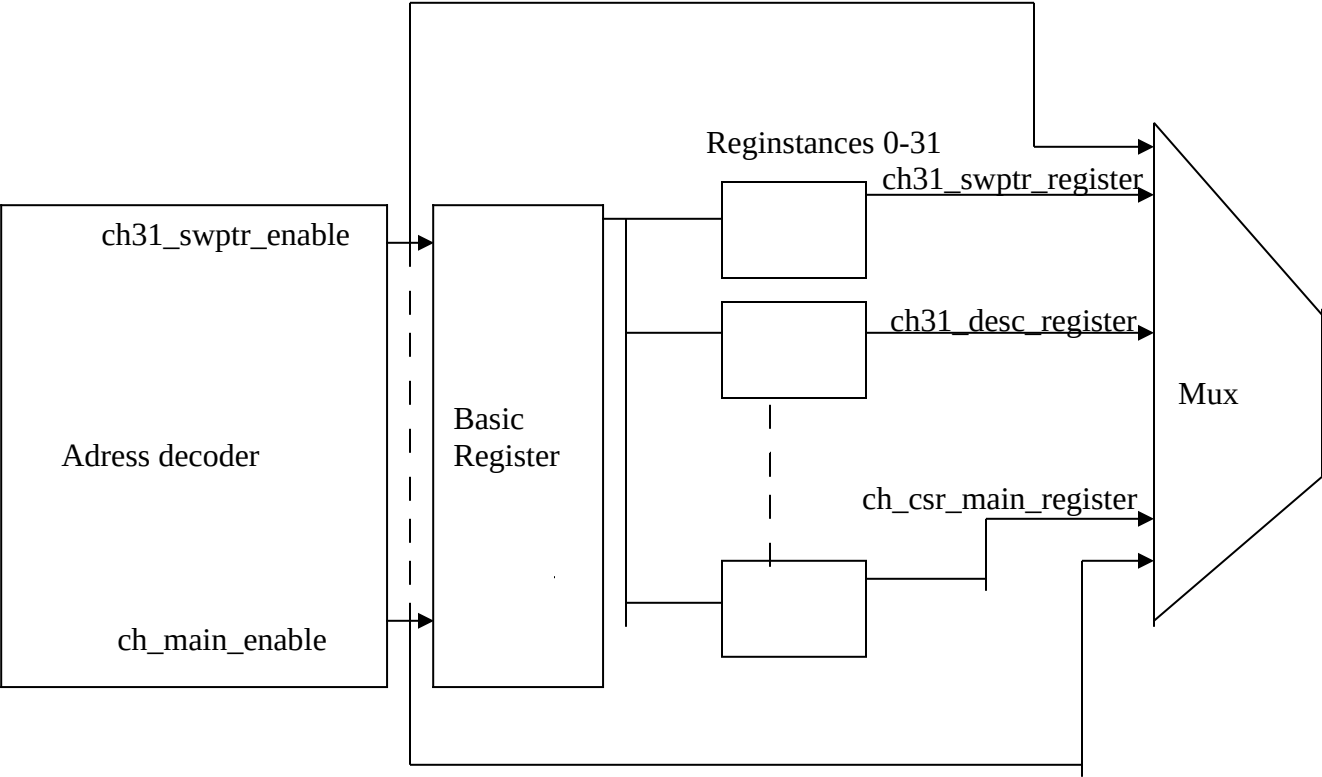
### **Multiplexer:**

The purpose of the multiplexer is to pass the data from the registers in case of read request.

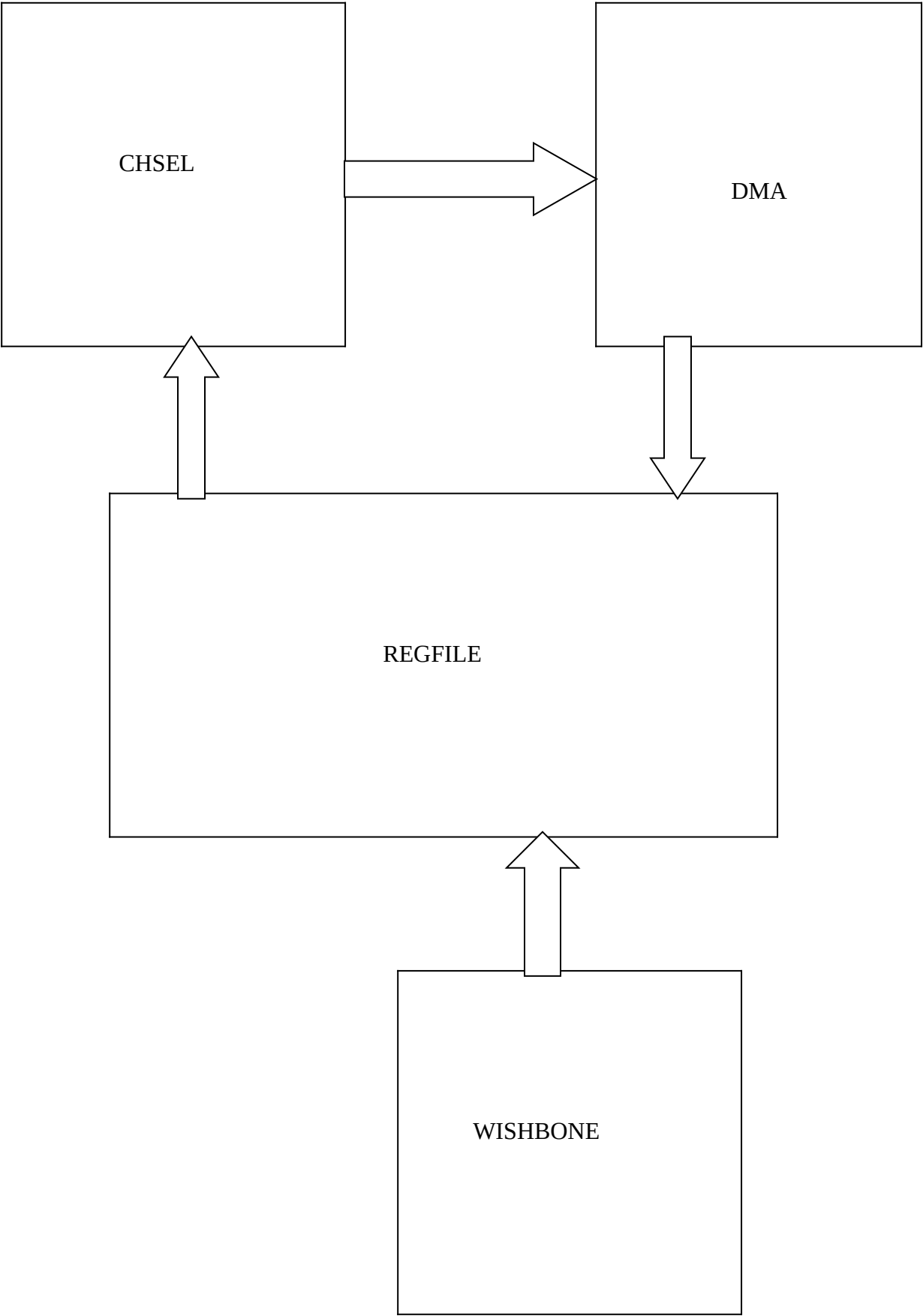
Select signals to the multiplexer are the address enable signals from the decoder. The 31,32 bit registers are the inputs to the multiplexer.

Based on the select signal, the data is read from that register. Once the data is read the multiplexer output is set back to its reset value.

Block Diagram



Flow Diagram



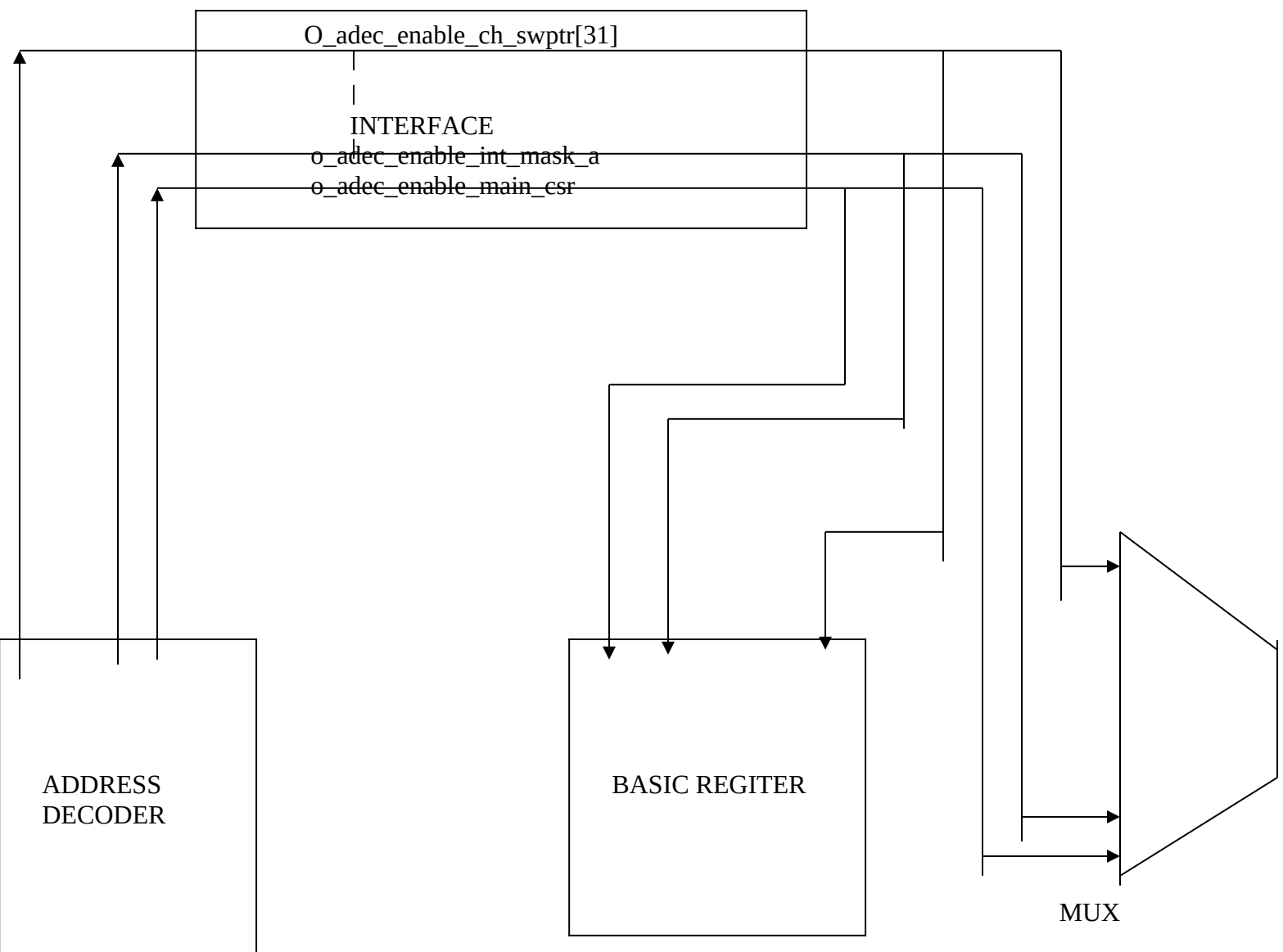
### **Register Interface:**

The regfile interface is the collection of the enable signals common to address decoder, basic register, multiplexer.

The interface serves to prevent the repetitive declarations of the enable ports common to three modules. The modport in the interface decides the direction of the port.

The Block diagram showing the interface connectivity in regfile follows

**Interface Connectivity**



**Adress decoder signals**  
=====

| Signal                       | width | direction |
|------------------------------|-------|-----------|
| i_clk                        | 1     | I         |
| i_rst                        | 1     | I         |
| i_addr                       | 32    | I         |
| i_wr                         | 1     | I         |
| i_rd                         | 1     | I         |
| i_wr_data                    | 32    | I         |
| O_adec_enable_main_csr       | 1     | O         |
| O_adec_enable_int_mask_a     | 1     | O         |
| O_adec_enable_int_mask_b     | 1     | O         |
| O_adec_enable_src_a          | 1     | O         |
| O_adec_enable_src_b          | 1     | O         |
| O_adec_enable_ch_csr[30:0]   | 1     | O         |
| O_adec_enable_ch_sz[30:0]    | 1     | O         |
| O_adec_enable_ch_ao[30:0]    | 1     | O         |
| O_adec_enable_ch_am0[30:0]   | 1     | O         |
| O_adec_enable_ch_a1[30:0]    | 1     | O         |
| O_adec_enable_ch_am1[30:0]   | 1     | O         |
| O_adec_enable_ch_desc{30:0}  | 1     | O         |
| O_adec_enable_ch_swptr[30:0] | 1     | O         |

1

### Regfileinstance signals.

=====

| Signal                   | width | direction |
|--------------------------|-------|-----------|
| i_clk                    | 1     | I         |
| i_rst                    | 1     | I         |
| i_wr                     | 1     | I         |
| i_rd                     | 1     | I         |
| i_wr_data                | 32    | I         |
| o_adec_enable_main_csr   | 1     | I         |
| o_adec_enable_int_mask_a | 1     | I         |
| o_adec_enable_int_mask_b | 1     | I         |

|                                   |    |   |
|-----------------------------------|----|---|
| o_adec_enable_src_a               | 1  | I |
| o_adec_enable_src_b               | 1  | I |
| o_adec_enable_ch_csr[30:0]        | 1  | I |
| o_adec_enable_ch_sz[30:0]         | 1  | I |
| o_adec_enable_ch_ao[30:0]         | 1  | I |
| o_adec_enable_ch_am0[30:0]        | 1  | I |
| o_adec_enable_ch_a1[30:0]         | 1  | I |
| o_adec_enable_ch_am1[30:0]        | 1  | I |
| o_adec_enable_ch_desc[30:0]       | 1  | I |
| o_adec_enable_ch_swptr[30:0]      | 1  | I |
| regfile_chsel_ch_main_csr[31:0]   | 32 | O |
| regfile_chsel_ch_int_mask_a[31:0] | 32 | O |
| regfile_chsel_ch_int_mask_b[31:0] | 32 | O |
| regfile_chsel_src_a[31:0]         | 32 | O |
| regfile_chsel_ch_src_b[31:0]      | 32 | O |
| regfile_chsel_chch_csr[31:0]      | 32 | O |
| regfile_chsel_ch_sz[31:0]         | 32 | O |
| regfile_chsel_ch_a0[31:0]         | 32 | O |
| regfile_chsel_ch_a1[31:0]         | 32 | O |
| regfile_chsel_ch_am1[31:0]        | 32 | O |
| regfile_chsel_ch_desc[31:0]       | 32 | O |
| regfile_chsel_ch_swptr [31:0]     | 32 | O |

### **Multiplexer signals**

=====

| Signal                   | width | direction |
|--------------------------|-------|-----------|
| i_clk                    | 1     | I         |
| i_rst                    | 1     | I         |
| i_wr                     | 1     | I         |
| i_rd                     | 1     | I         |
| i_wr_data                | 32    | I         |
| o_adec_enable_main_csr   | 1     | I         |
| o_adec_enable_int_mask_a | 1     | I         |
| o_adec_enable_int_mask_b | 1     | I         |

|                              |    |   |
|------------------------------|----|---|
| o_adec_enable_src_a          | 1  | I |
| o_adec_enable_src_b          | 1  | I |
| o_adec_enable_ch_csr[31:0]   | 1  | I |
| o_adec_enable_ch_sz[31:0]    | 1  | I |
| o_adec_enable_ch_ao[31:0]    | 1  | I |
| o_adec_enable_ch_am0[31:0]   | 1  | I |
| o_adec_enable_ch_a1[31:0]    | 1  | I |
| o_adec_enable_ch_am1[31:0]   | 1  | I |
| o_adec_enable_ch_desc[31:0]  | 1  | I |
| o_adec_enable_ch_swptr[31:0] | 1  | I |
| regfile_wb_data              | 32 | O |